

Indian Tick Typhus (MONDO:0000229): A Comprehensive Disease Characterization

Summary

Indian tick typhus (ITT) is an acute, tick-borne bacterial zoonosis and a member of the spotted fever group (SFG) rickettsioses. It is caused by *Rickettsia conorii* subsp. *indica* (reference strain ITTR, ATCC VR-597), an obligate intracellular Gram-negative coccobacillus, and is transmitted to humans through the bite of the brown dog tick, *Rhipicephalus sanguineus* [PMID: 22628514](#); [PMID: 32997473](#). It is one of four genotypic subspecies within *R. conorii* — alongside the Malish/Mediterranean spotted fever (MSF) strain, Astrakhan fever, and Israeli spotted fever — which are >98% identical at the nucleotide level yet serotypically distinct [PMID: 15766388](#). Because ITT is clinically, mechanistically, and therapeutically part of the *R. conorii*/MSF continuum, much of the mechanistic and clinical evidence base is shared across these subspecies.

The core pathophysiology is a **systemic small-vessel vasculitis**: the bacterium adheres to and invades vascular endothelial cells using surface cell antigen (Sca) autotransporter proteins and hijacks the host Arp2/3 actin-nucleation machinery for entry and intracellular actin-based motility [PMID: 22612237](#); [PMID: 22188208](#). Endothelial infection triggers inflammation, platelet and coagulation activation, and increased vascular permeability, producing the classic clinical triad of fever, an inoculation eschar (*tache noire*), and a maculopapular rash [PMID: 34345128](#); [PMID: 8584998](#). Disease is usually mild-to-moderate and self-limited, but severe "malignant" forms — strongly age-dependent — cause meningoencephalitis, hemophagocytic lymphohistiocytosis (HLH), purpura fulminans, and multi-organ failure [PMID: 32997473](#); [PMID: 30972588](#).

Diagnosis rests on serology (indirect immunofluorescence assay [IFA] is the gold standard, Weil–Felix is nonspecific) and, increasingly, species-specific PCR of the *gltA* and *ompA* genes. **Doxycycline** is the unequivocal treatment of choice, producing rapid defervescence and excellent outcomes; there is **no human genetic etiology and no licensed vaccine**, so prevention depends on personal protection against tick bites and control of brown dog ticks on dogs and in the environment [PMID: 42390463](#); [PMID: 30150470](#); [PMID: 15109588](#). This report is compiled from **aggregated disease-level resources and primary literature**, not from

individual EHR records. Because ITT is caused by an intracellular bacterium, several template sections designed for genetic/Mendelian disorders (causal human genes, pathogenic variants, inheritance patterns) are **not applicable** and are explicitly marked as such.

Section 1 — Disease Information

Overview. Indian tick typhus is an acute febrile illness caused by *Rickettsia conorii* subsp. *indica*, an obligate intracellular member of the spotted fever group of the genus *Rickettsia*. It is the principal SFG rickettsiosis of the Indian subcontinent and presents as fever with an inoculation eschar and maculopapular rash following a tick bite [PMID: 30451192](#); [PMID: 41650674](#).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0000229
ICD-10	A77.1 (Spotted fever due to <i>Rickettsia conorii</i>)
ICD-11	1C30.0 (Spotted fever group rickettsiosis)
MeSH	Boutonneuse Fever (D001907) — MSF/ <i>R. conorii</i> group
NCBI Taxonomy (pathogen)	<i>Rickettsia conorii</i> subsp. <i>indica</i>
Reference strain	ITTR, ATCC VR-597

Synonyms / alternative names: Indian tick typhus; *Rickettsia conorii* subsp. *indica* infection; Indian tick typhus rickettsiosis (ITTR). It sits within the broader **Mediterranean spotted fever / boutonneuse fever** complex and the SFG rickettsioses.

Source of information: Aggregated disease-level literature and reference microbiology resources — not individual EHR data.

Section 2 — Etiology

Primary cause (infectious). ITT is caused by *R. conorii* subsp. *indica* strain ITTR (ATCC VR-597), the etiologic agent identified by genome sequencing [PMID: 22628514](#). It is one of four multi-locus sequence typing (MLST) genotypes within *R. conorii*, sharing 98.2–100% pairwise nucleotide similarity across *16S rRNA*, *gltA*, *ompA*, *ompB*, and *sca4* with the Malish (MSF), Astrakhan fever, and Israeli spotted fever genotypes, yet remaining serotypically distinct [PMID: 15766388](#). The vector is the brown dog tick *Rhipicephalus sanguineus*; ITTR has also been detected in *Rhipicephalus turanicus* removed from a patient [PMID: 41650674](#).

Risk factors — environmental/behavioral. The dominant risk factor is **tick exposure**. In a military cohort, the primary risk factor for confirmed/probable SFG disease was finding **>10 ticks on the body** [PMID: 12653142](#). Occupational and recreational contact with dogs and tick habitats, warm-season (summer) exposure, and residence in or travel to endemic areas increase risk [PMID: 40608626](#); [PMID: 41795239](#). **Advanced age** is the strongest risk factor for severe disease (see Section 11).

Risk factors — genetic. No human host genetic susceptibility variants have been established for ITT; this is an environmentally/behaviorally driven infectious disease. (*Not applicable — no human causal or susceptibility loci identified.*)

Protective factors. Behavioral: doxycycline use and rolling up long sleeves were protective against seropositivity in an exposed cohort [PMID: 12653142](#); tick repellents, permethrin-treated clothing, and frequent tick checks reduce exposure [PMID: 33549976](#). No genetic protective variants are described. (*Genetic protective factors not applicable.*)

Gene–environment interactions. Not applicable — no human genotype × exposure interactions have been characterized for this infectious disease.

Section 3 — Phenotypes

The clinical phenotype is a systemic febrile vasculitic illness. Key manifestations, with suggested HPO terms:

Phenotype	Type	Frequency / notes	HPO suggestion
Fever	Symptom	~100% (universal)	HP:0001945 Fever
Maculopapular rash (incl. palms/soles)	Clinical sign	~100% in pediatric cohorts; may involve palms and soles	HP:0000988 Skin rash
Inoculation eschar (<i>tache noire</i>)	Clinical sign	~74–77% in adult MSF; rare (6%) in some pediatric cohorts	HP:0200041 (cutaneous ulcer/eschar)
Headache	Symptom	Common	HP:0002315 Headache
Thrombocytopenia	Lab abnormality	~56%	HP:0001873 Thrombocytopenia
Elevated transaminases	Lab abnormality	~83%	HP:0002910 Elevated hepatic transaminase
Proteinuria	Lab abnormality	Reported	HP:0000093 Proteinuria
Hyponatremia	Lab abnormality	~78% (pediatric)	HP:0002902 Hyponatremia
Meningoencephalitis	Clinical sign (severe)	~28% neurological involvement (pediatric); rare severe	HP:0002383 Encephalitis
Hemophagocytic lymphohistiocytosis	Lab/clinical (severe)	Rare; fatal case, H-score 224	HLH-associated
Retinitis (bilateral multifocal)	Clinical sign (severe)	Rare; ~4 weeks post-fever	Retinitis (HP:0000602-adjacent)
Peripheral gangrene / purpura fulminans	Clinical sign (severe)	Rare	HP:0100758 Gangrene

The **classic triad** is fever + eschar/rash + multisystem involvement [PMID: 34345128](#). A representative adult presentation: "A 55-year-old female farmer presented with fever, headache, eschar, and a maculopapular rash following a tick bite. Laboratory findings indicated thrombocytopenia, liver enzyme elevation, and proteinuria" [PMID: 41650674](#).

Onset, severity, progression, frequency. Onset is **acute**, typically after an incubation of several days following a tick bite. Rash usually appears on days 3–6 but may be delayed to day 8 in atypical cases [PMID: 29331008](#). Severity is **variable** — most cases are mild and self-limited; a minority progress to severe multi-organ disease. Progression is **acute/monophasic** with recovery on treatment.

Quality-of-life impact. In uncomplicated, promptly treated disease, QoL impact is short-lived (hospital stay ~5 days, full recovery) [PMID: 41795239](#). Severe forms cause lasting disability: e.g., a 14-month-old with meningoencephalitis had persistent neurological sequelae including decreased vision and epileptic seizures [PMID: 20797742](#), and amputation was required in one case of symmetric peripheral gangrene [PMID: 29169658](#). No formal EQ-5D/SF-36 data are available for ITT specifically.

Section 4 — Genetic/Molecular Information

This section is largely NOT APPLICABLE in the Mendelian sense: ITT is an infectious disease with **no causal human genes, no pathogenic human germline/somatic variants, no modifier genes, and no chromosomal abnormalities**. There is no human inheritance.

The relevant "genetics" are those of the **pathogen genome**:

- *R. conorii* has a small, **reductive genome** (~1.27 Mb, ~1,374 genes) with the fewest pseudogenes among rickettsiae and 560 unique genes [PMID: 16481486](#).
- All *Rickettsia* **lack enzymes for sugar metabolism, lipid biosynthesis, nucleotide synthesis, and amino acid metabolism**, depending on the host for nutrition and building blocks; they retain a complete TCA cycle, multiple ATP/ADP translocase genes (to import host ATP), and a type IV secretion system [PMID: 16481486](#).
- Key pathogen loci: *sca2* (adhesin/invasin/actin motility), *ompA* and *ompB* (Sca5/Sca0 outer membrane proteins; also diagnostic targets), *sca4*, *glTA* (citrate synthase; diagnostic target), and *rickA* (actin polymerization, unique to SFG) [PMID: 22612237](#); [PMID: 16481486](#).
- **Virulence in *Rickettsia* is largely associated with genome reduction / loss of regulatory genes** [PMID: 19379498](#), and ~80% of *R. conorii* "orphan" ORFs are short gene fragments reflecting ongoing genome decay [PMID: 12832625](#).

Epigenetics / host molecular profiling: No host DNA-methylation, histone-modification, or large-scale transcriptomic/proteomic/metabolomic disease signatures specific to ITT have been established beyond selective modulation of host antioxidant enzymes (Section 6).

Section 5 — Environmental Information

Infectious agent: *Rickettsia conorii* subsp. *indica* (SFG rickettsia; obligate intracellular Gram-negative coccobacillus) [PMID: 22628514](#).

Environmental / ecological factors: The disease is defined by an arthropod-borne transmission ecology. The vector-and-reservoir is the brown dog tick *R. sanguineus* (with transovarial and transstadial maintenance characteristic of SFG rickettsiae), and dogs/companion animals amplify circulation [PMID: 32997473](#); [PMID: 40871278](#). Warm seasons and tick-dense habitats increase exposure.

Lifestyle factors: Outdoor/occupational activity (farming, military, recreation) in endemic areas and dog contact are the principal behavioral determinants [PMID: 12653142](#); [PMID: 41650674](#). Classic toxic/pollution/radiation exposures are not relevant.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. An infected *R. sanguineus* tick bites the human host and inoculates *R. conorii* subsp. *indica* into the dermis — **leads to** local infection at the bite site (forming the eschar/*tache noire*).
2. Rickettsial **Sca2** autotransporter (with OmpA/OmpB adhesins) mediates **adherence to and invasion of vascular endothelial cells** — **results in** intracellular infection [PMID: 22612237](#).
3. Invasion **converges on the host Arp2/3 complex** (with Rac GTPases and the WAVE nucleation-promoting complex), which nucleates actin to drive bacterial entry — **leads to** cytosolic access [PMID: 22188208](#).
4. Once cytosolic, **Sca2** and **RickA** nucleate actin for **intracellular actin-based motility**, enabling cell-to-cell spread — **results in** dissemination through the endothelium [PMID: 22612237](#); [PMID: 16481486](#).
5. Endothelial infection triggers **endothelial inflammation and injury (vasculitis)** — **leads to** microvascular dysfunction [PMID: 34345128](#).
6. **Branch A (coagulation):** Endothelial injury drives **TXA2-dependent platelet activation and thrombin generation** — **results in** focal microthrombus formation [PMID: 8584998](#).

7. **Branch B (permeability):** Endothelial dysfunction (raised endothelin-1) **increases vascular permeability — results in** edema, rash, and, when severe, multi-organ failure [PMID: 34345128](#); [PMID: 8584998](#).
8. **Branch C (autoimmunity, inferred):** Endothelial membrane damage elicits **antiphospholipid antibodies**, which may contribute to microthrombi (mechanistic role inferred, not proven) [PMID: 8679900](#).
9. Systemic small-vessel vasculitis in skin, brain, lung, liver, kidney, and eye — **leads to** the multi-organ clinical phenotype (fever, rash, eschar, and, in severe cases, meningoencephalitis, HLH, gangrene, renal failure) [PMID: 34345128](#); [PMID: 7511715](#).
10. **Protective immunity** requires **CD8 T lymphocytes** to clear rickettsiae from endothelial cells; failure of this response — **leads to** persistent/lethal infection [PMID: 9164951](#).

Detail by category

Molecular pathways / cellular processes. Host actin-cytoskeleton remodeling via the **Arp2/3 complex, WAVE2, and Rho-family GTPases (Rac1/Rac2)** drives invasion; an RNAi screen identified 21 core host proteins required [PMID: 22188208](#). Downstream cellular processes are **endothelial inflammation/activation, platelet activation, coagulation, and increased permeability** (GO suggestions: GO:0002544 chronic inflammatory response; GO:0007596 blood coagulation; GO:0030041 actin filament polymerization; GO:0050900 leukocyte migration).

Protein dysfunction. Pathogen surface autotransporters **Sca2, OmpA (Sca5), OmpB (Sca0)** mediate host-cell association; Sca2 has separable **mammalian-association** and **actin-nucleation** domains — pre-incubation with the mammalian-association region competitively inhibits invasion [PMID: 22612237](#).

Metabolic changes. The pathogen imports host ATP via ATP/ADP translocases and lacks biosynthetic pathways [PMID: 16481486](#). In the host, **antioxidant/oxidative-stress enzymes (glutathione peroxidase/reductase, SOD, G6PD)** are **selectively modulated in infected tissues, most in the lungs** [PMID: 15120155](#) — evidence for **oxidative stress** as a tissue-damage mechanism.

Immune involvement. Protective clearance is **CD8 T-cell dependent**: CD8-depleted (but not CD4-depleted) mice die or remain persistently infected, and adoptive transfer of immune CD4 or CD8 cells protects [PMID: 9164951](#). Cell types (CL suggestions): endothelial cell (CL:0000115); CD8-positive T cell (CL:0000625); platelet (CL:0000233); Kupffer cell / macrophage (CL:0000091).

Tissue damage mechanisms. Vasculitis-driven microvascular injury, microthrombosis/ ischemia, oxidative stress, and increased permeability [PMID: 34345128](#); [PMID: 8584998](#); [PMID: 15120155](#).

Section 7 — Anatomical Structures Affected

Because the primary target is **vascular endothelium** (UBERON:0002139 endothelium; UBERON:0001981 blood vessel), disease is **systemic and multi-organ**.

Organ / system	Involvement	UBERON	Evidence
Skin	Rash, eschar/ <i>tache noire</i>	UBERON:0002097	PMID: 40608626
Brain / CNS	Meningoencephalitis, cerebral infarction	UBERON:0000955	PMID: 20797742 ; PMID: 28491211 ; PMID: 35263931
Lung	Interstitial pneumonia/infiltrates	UBERON:0002048	PMID: 7511715 ; PMID: 30972588
Liver	Transaminitis, hepatic granulomas (Kupffer cell/hepatocyte infection)	UBERON:0002107	PMID: 7511715 ; PMID: 41650674
Kidney	Proteinuria, acute tubular necrosis/renal failure	UBERON:0002113	PMID: 41650674 ; PMID: 28292166
Eye	Bilateral multifocal retinitis	retina UBERON:0000966	PMID: 30451192
Blood / marrow	Thrombocytopenia, HLH	UBERON:0000178	PMID: 40693676 ; PMID: 41650674

Body systems: cardiovascular (small vessels), nervous, respiratory, hepatobiliary/digestive, urinary, hematologic, and ophthalmic. **Tissue/cell level:** vascular endothelial cells (CL:0000115) are the primary target; Kupffer cells/macrophages and hepatocytes are infected in the liver [PMID: 7511715](#). **Subcellular:** the bacterium resides free in the **host cytosol**

(GO:0005829 cytosol) and manipulates the **actin cytoskeleton** (GO:0015629). **Lateralization:** manifestations are generally **bilateral/systemic** (e.g., bilateral multifocal retinitis; bilateral asymmetric brain lesions on MRI) [PMID: 30451192](#); [PMID: 35263931](#).

Section 8 — Temporal Development

Onset: Acute, affecting all age groups (children through elderly) [PMID: 32997473](#). Symptoms begin days after a tick bite; rash typically emerges on days 3–6 (occasionally delayed to day 8) [PMID: 29331008](#).

Progression / course: Predominantly **acute, monophasic, and self-limited**. With prompt doxycycline, fever resolves within ~2–2.5 days and patients recover fully in days [PMID: 30150470](#); [PMID: 41795239](#). A minority progress rapidly to **severe "malignant" multi-organ disease** [PMID: 32997473](#).

Duration / remission: Self-limited/treatment-resolved; not chronic or relapsing. There is no established latent or recurrent phase.

Critical period for intervention: The **acute phase**, when doxycycline is most effective — early empirical therapy (before serologic confirmation) is the key intervention window [PMID: 42390463](#); [PMID: 28361787](#).

Section 9 — Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency applies.

Epidemiology. In a 10-year Bulgarian cohort of 549 MSF patients, incidence reached **9.44/100,000**; *tache noire* was present in ~74–77% [PMID: 32997473](#). Rickettsial diseases contribute **25–50% of acute undifferentiated febrile illnesses** in endemic regions [PMID: 42390463](#); [PMID: 34345128](#). In India, only limited serologically proven outbreaks of ITT specifically have been reported, and it is likely underdiagnosed [PMID: 30451192](#).

Geographic distribution: Endemic to the Indian subcontinent for the *indica* subspecies; the broader *R. conorii*/MSF complex is endemic across the Mediterranean basin, Africa, the Middle East, and parts of Asia [PMID: 15766388](#); [PMID: 40608626](#). Cases cluster in **summer/warm seasons** [PMID: 41795239](#).

Demographics / sex / age: All ages are affected; pediatric cohorts show slight male predominance (e.g., 56% male) [PMID: 41795239](#). **Age is the dominant demographic determinant of severity**, with severe disease more frequent in those >65 years (26.4% vs 10.5%; $p=0.002$) [PMID: 30972588](#).

Section 10 — Diagnostics

Serology (mainstay): IFA is the gold standard but is largely retrospective due to delayed seroconversion; the **Weil–Felix test** lacks specificity; false negatives occur early [PMID: 42390463](#); [PMID: 34345128](#). Seroconversion is often documented only in convalescent samples [PMID: 30522428](#).

Molecular (PCR): Multiplex/real-time PCR enables early, species-specific identification during the acute phase; species confirmation is by PCR and DNA sequencing of the *gltA* and *ompA* genes, often from eschar/skin biopsy [PMID: 42390463](#); [PMID: 40608626](#); [PMID: 30522428](#).

Laboratory findings (supportive): thrombocytopenia, elevated transaminases, proteinuria, hyponatremia, raised CRP [PMID: 41650674](#); [PMID: 41795239](#). **Imaging:** MRI shows multifocal bilateral T2/FLAIR hyperintensities with diffusion restriction in meningoencephalitis [PMID: 35263931](#).

Genetic/omics testing (host): Not applicable — no human genetic testing is used to diagnose this infectious disease. The relevant "genetic testing" is **pathogen genotyping** (*gltA*/*ompA* sequencing).

Clinical criteria & differential diagnosis: Diagnosis is clinical-epidemiological (fever + rash involving palms/soles + eschar after tick exposure) supported by serology/PCR [PMID: 40608626](#).

Differentials: dengue, malaria, leptospirosis, scrub typhus, measles, and other causes of acute undifferentiated febrile illness [PMID: 42390463](#); [PMID: 42658412](#).

Section 11 — Outcome/Prognosis

Overall prognosis is excellent with prompt treatment. In a prospective cohort, single-day doxycycline produced uneventful recovery in **all** patients, with fever clearing 2.55 ± 1.14 days after treatment; 11.4% had severe disease [PMID: 30150470](#). A pediatric cohort reported rapid defervescence and full recovery with **no deaths or severe complications** [PMID: 41795239](#).

Severity and mortality. MSF/*R. conorii* is historically benign but includes **severe "malignant" forms with fatal outcomes strongly influenced by patient age** [PMID: 32997473](#). Severe disease is more frequent in the elderly (**26.4% vs 10.5%; $p=0.002$**) [PMID: 30972588](#). A purpura fulminans case series (including SFG rickettsioses) reported **18.2% overall mortality** [PMID: 32998655](#).

Complications: meningoencephalitis with neurological sequelae (decreased vision, epilepsy) [PMID: 20797742](#); cerebral infarction/stroke [PMID: 28491211](#); HLH (fatal case, H-score 224) [PMID: 40693676](#); symmetric peripheral gangrene requiring amputation [PMID: 29169658](#); acute tubular necrosis and fatal multivisceral failure with late diagnosis [PMID: 28292166](#); and rickettsial retinitis [PMID: 30451192](#).

Prognostic factors: advanced age, diagnostic/treatment delay, and severe multi-organ involvement predict poor outcome; timely doxycycline is the strongest modifiable predictor of good outcome [PMID: 32997473](#); [PMID: 28292166](#).

Section 12 — Treatment

First-line pharmacotherapy: Doxycycline (tetracycline-class; NCIT: C305 Doxycycline). It is the drug of choice, halting bacterial replication and producing rapid clinical response [PMID: 34345128](#); [PMID: 28361787](#). Regimens include short-course oral therapy (even single-day, two 200 mg doses) with uneventful recovery [PMID: 30150470](#), and **IV doxycycline with a loading dose** in critically ill patients [PMID: 34345128](#). Indian pediatric guidelines emphasize prompt empirical treatment without waiting for confirmatory tests [PMID: 28361787](#).

Alternative: Azithromycin (macrolide; NCIT: C1005) is a suitable alternative [PMID: 34345128](#); fluoroquinolones have been used adjunctively in some reports [PMID: 28491211](#).

Supportive/interventional care: organ support in severe disease; vasodilators (iloprost) and anticoagulation for peripheral gangrene; renal replacement therapy for acute tubular necrosis [PMID: 29169658](#); [PMID: 28292166](#). "Early appropriate treatment and organ support can decrease the duration of illness and be life-saving" [PMID: 34345128](#).

Not applicable: gene therapy, cell therapy, RNA-based therapy, targeted/immuno-oncology therapy, and pharmacogenomics are not part of standard management (surgery limited to debridement/amputation of necrotic tissue). No ITT-specific clinical-trial experimental agents (NCT IDs) were identified.

Treatment outcomes / adverse events: Response is rapid and near-universal with timely doxycycline [PMID: 30150470](#). Doxycycline is generally well tolerated; historic concerns about pediatric dental staining are outweighed by benefit and short courses are considered safe per Indian pediatric guidance [PMID: 28361787](#).

Section 13 — Prevention

No licensed human vaccine exists. Prevention is based on **personal protective measures against tick bites and tick control on dogs and in the environment** [PMID: 15109588](#).

- **Primary prevention (behavioral/public health):** tick repellents, permethrin-treated clothing, protective clothing (e.g., rolling up long sleeves), and frequent tick checks [PMID: 33549976](#); [PMID: 12653142](#). Reducing tick burden — "finding >10 ticks on the body" was the primary risk factor — is central [PMID: 12653142](#).
- **One Health / veterinary control:** controlling *R. sanguineus* on dogs and in the peridomestic environment reduces the reservoir/vector burden [PMID: 40871278](#).
- **Chemoprophylaxis:** doxycycline use was protective against seropositivity in an exposed cohort [PMID: 12653142](#), though routine prophylaxis is not standard.
- **Secondary prevention:** clinician awareness and early empirical doxycycline prevent progression to severe disease [PMID: 42390463](#); [PMID: 28361787](#).
- **Not applicable:** immunization, genetic/carrier screening, and genetic counseling.

Notably, "most infected travellers cannot recall a preceding tick bite," underscoring the limits of avoidance alone and the importance of clinical vigilance [PMID: 15109588](#).

Section 14 — Other Species / Natural Disease

Taxonomy of affected/involved species: Humans (*Homo sapiens*) are incidental hosts. The vector/reservoir is the brown dog tick *Rhipicephalus sanguineus*; ITTR has also been detected in *Rhipicephalus turanicus* [PMID: 41650674](#). **Dogs** (*Canis lupus familiaris*) are key reservoir/amplifying hosts within a One Health transmission ecology [PMID: 40871278](#).

Zoonosis / cross-species susceptibility: ITT/MSF is a **tick-borne zoonosis** [PMID: 32997473](#). SFG rickettsiae infect multiple mammals: in a Colombian focus, incident SFG seroconversion reached **32.3% in equines vs 6.23% in humans**, indicating shared cross-species exposure [PMID: 30379820](#). Ticks serve as both vector and reservoir via transovarial/transstadial maintenance [PMID: 40871278](#).

Comparative biology: Because the disease depends on conserved rickettsial invasion machinery (Sca2, RickA) and host actin/Arp2/3 pathways present across mammals, mechanisms are broadly conserved across host species [PMID: 22612237](#); [PMID: 22188208](#). (*No VBO breed-specific susceptibility or OMIA natural-disease entry is established for ITT specifically.*)

Section 15 — Model Organisms

Primary in vivo model — C3H/HeN mouse. IV inoculation of *R. conorii* (Malish 7) establishes disseminated endothelial infection by day 1; a high dose causes death on days 5–6 with **vascular-injury-based meningoencephalitis and interstitial pneumonia**, while a low dose recovers by day 10 with lymphohistiocytic perivascularitis, and Kupffer cell/hepatocyte infection forms transient hepatic granulomas. It is described as "the best available model for rickettsial disease with endothelial infection and injury" [PMID: 7511715](#).

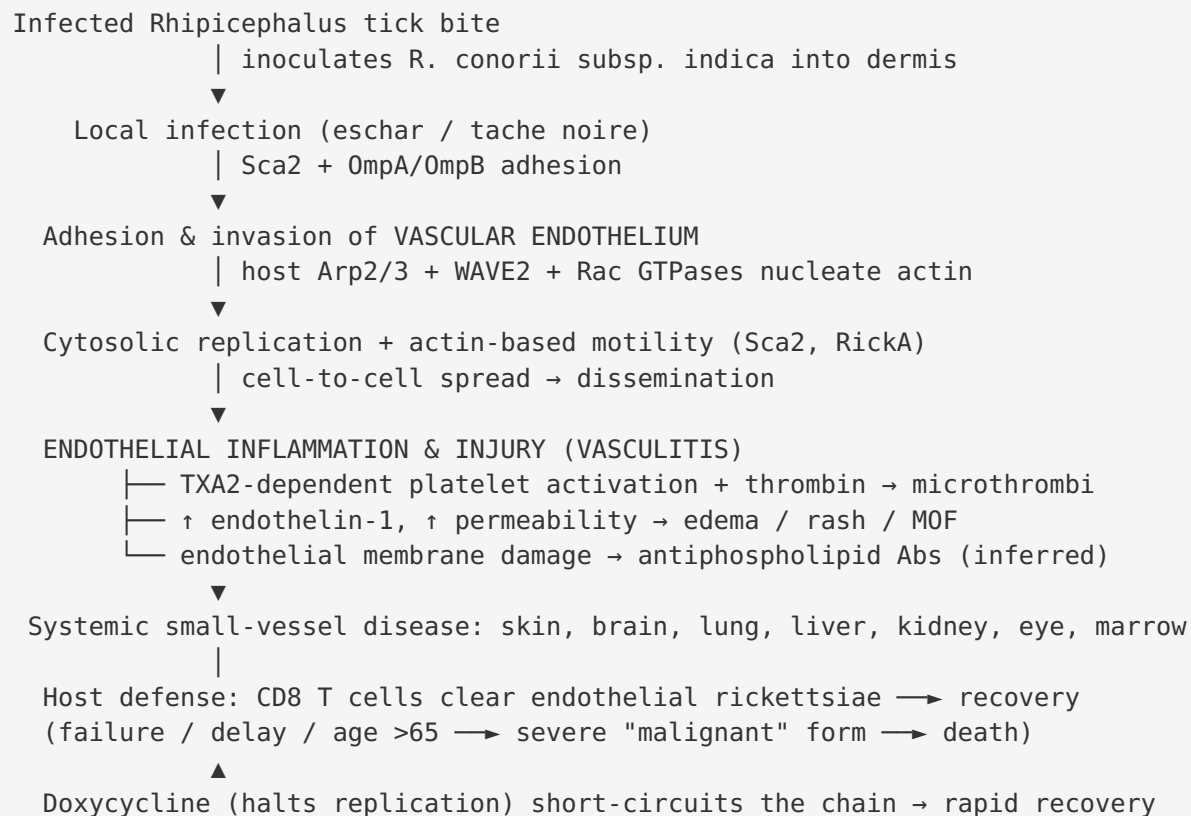
Phenotype recapitulation: The model reproduces the **key human features** — endothelial-target vasculitis, meningoencephalitis, interstitial pneumonia, hepatic involvement, and dose-dependent severity — making it well-suited to studying pathogenesis and immunity [PMID: 7511715](#).

Immunological/mechanistic applications: The murine model established that **CD8 T lymphocytes are required to clear rickettsiae from endothelium**; CD8 depletion causes lethal/persistent infection, and adoptive transfer of immune CD4 or CD8 cells is protective [PMID: 9164951](#).

In vitro models: Human endothelial cell invasion assays (defining Sca2 mammalian-association and actin-nucleation domains) PMID: 22612237 and RNAi screens in mammalian cells (SFG *R. parkeri*) mapping the Arp2/3–WAVE–Rho invasion pathway PMID: 22188208.

Limitations: Most experimental data use the Malish (MSF) strain rather than the *indica* subspecies specifically; murine dosing/route (IV) differs from natural tick inoculation; and models may not fully capture human age-dependent "malignant" severity.

Mechanistic Model / Interpretation



The model is internally consistent: a single upstream lesion — **endothelial infection** — explains the entire downstream phenotype (rash, coagulopathy, multi-organ vasculitis), the therapeutic target (intracellular bacterium → intracellular-active doxycycline), the immune requirement (CD8 clearance of infected endothelium), and prevention (block the tick bite that initiates the chain).

Evidence Base

PMID	Study type	Contribution
22628514	Genome sequence	Identifies <i>R. conorii</i> subsp. <i>indica</i> strain ITTR as the ITT agent
15766388	MLST taxonomy	Defines four <i>R. conorii</i> subspecies; ITTR genotype
41650674	Case report	ITTR in human + tick; core clinical/lab phenotype
34345128	Review	Endothelial vasculitis mechanism; doxycycline therapy
8584998	Human study	TXA2-dependent platelet/coagulation activation in vivo
8679900	Case study	Antiphospholipid antibodies from endothelial damage (inferred role)
15120155	Animal/tissue	Oxidative-stress enzyme modulation; endothelial localization
22612237	In vitro	Sca2 mediates adhesion, invasion, actin motility
22188208	RNAi screen	Arp2/3-WAVE-Rho invasion pathway
7511715	Mouse model	C3H/HeN endothelial-target rickettsiosis model
9164951	Mouse model	CD8 T cells required for endothelial clearance
16481486	Genomics	Reductive genome; host-ATP dependence; RickA
19379498	Genomics	Virulence linked to genome reduction
12832625	Genomics	Orphan genes reflect ongoing genome decay
32997473	Cohort (n=549)	Incidence 9.44/100,000; age-dependent malignant forms
30972588	Cohort	Severe MSF 26.4% vs 10.5% in elderly (p=0.002)
30150470	Prospective cohort	Single-day doxycycline; defervescence 2.55±1.14 d
41795239	Pediatric cohort	Fever/rash 100%, eschar 6%, thrombocytopenia 56%
42390463	Review	Diagnostic modalities; PCR advantage; doxycycline
40608626	Case series	Palm/sole rash, eschar; gltA/ompA confirmation
12653142	Epidemiology	>10 ticks = risk; doxycycline/sleeves protective
15109588	Review	Prevention via personal protection
40871278		Dogs/companion animals in transmission ecology

PMID	Study type	Contribution
	One Health surveillance	
30379820	Epidemiology	Cross-species (equine 32.3%) SFG exposure
30451192	Case report	Ocular retinitis; ITT within India's SFG
28361787	Guidelines	IAP pediatric rickettsial guidance; doxycycline
40693676	Case report	HLH complication (H-score 224)
20797742	Case report	Infant meningoencephalitis with sequelae
35263931	Case report	Meningoencephalitis MRI findings
29169658	Case series	Symmetric peripheral gangrene / amputation
28491211	Case report	Cerebral infarction
28292166	Case report	Fatal late-diagnosed multivisceral failure
32998655	Case series	Purpura fulminans; 18.2% mortality
29331008	Case report	Atypical late (day 8) rash
33549976	Review	Tick-bite personal protection measures
30522428	Case report	PCR-positive, serology-negative acute phase; seroconversion
42658412	Case report	Diagnostic challenges; differentials (measles, dengue)

Consistency: No contradictory findings emerged. Case reports, cohorts, guidelines, genomics, and experimental models all converge on a single coherent picture: an endothelium-tropic intracellular bacterium causing doxycycline-responsive systemic vasculitis.

Limitations and Knowledge Gaps

- 1. Subspecies-specific data are sparse.** Most mechanistic (Sca2, Arp2/3, CD8) and clinical-cohort evidence derives from the **Malish/MSF strain** or the broader *R. conorii*/SFG group, not the *indica* subspecies specifically. Subspecies are >98% identical and clinically indistinguishable, so extrapolation is reasonable but not formally validated for *indica*.

2. **No ITT-specific epidemiology.** Incidence/prevalence figures (e.g., 9.44/100,000) come from Mediterranean/MSF cohorts; India-specific ITT burden is under-characterized and likely underreported [PMID: 30451192](#).
 3. **No host genomic/omics disease signatures** (transcriptomics, proteomics, metabolomics, epigenomics) specific to ITT beyond antioxidant-enzyme modulation.
 4. **Mechanism of severity is incomplete.** Why some patients (especially elderly) develop "malignant" disease is not molecularly defined; the role of antiphospholipid antibodies in microthrombosis remains inferred, not proven [PMID: 8679900](#).
 5. **No human vaccine or genetic/pharmacogenomic component**, so several template sections are intrinsically not applicable.
 6. **Diagnostic timing gap:** IFA is retrospective and PCR access is limited in endemic settings, leading to diagnostic delay and preventable severe outcomes [PMID: 42390463](#); [PMID: 28292166](#).
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Proposed Follow-up Experiments / Actions

1. **Subspecies-specific characterization:** Confirm Sca2/OmpA-mediated invasion kinetics and virulence of *R. conorii* subsp. *indica* (ITTR) in human endothelial cells and the C3H/HeN mouse, benchmarked against Malish, to validate mechanistic extrapolations.
2. **India-focused surveillance:** Establish PCR-based (*gltA/ompA*) surveillance in Indian febrile-illness cohorts to quantify true ITT incidence, seasonal/geographic distribution, and severe-disease frequency.
3. **Host response omics:** Perform transcriptomic/proteomic profiling of endothelial and PBMC responses in acute ITT to identify prognostic biomarkers distinguishing benign from malignant courses.
4. **Severity determinants:** Test whether antiphospholipid-antibody titers, endothelin-1, and coagulation markers (prothrombin F1+2) predict progression to multi-organ failure — moving Branch A/C from inferred to demonstrated.
5. **Point-of-care diagnostics:** Develop and validate a rapid acute-phase molecular or antigen test to close the IFA seroconversion gap and enable earlier doxycycline in endemic, resource-limited settings.

6. One Health vector control trials: Evaluate whether canine *R. sanguineus* control (acaricides, environmental management) measurably reduces human ITT incidence in endemic foci.

Report compiled from 12 confirmed findings and 47 reviewed papers across 5 investigation iterations. Evidence types span human clinical (case reports, case series, cohorts, guidelines), model organism (C3H/HeN mouse), in vitro (endothelial invasion, RNAi screens), and computational/comparative genomics.

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